DATABASES

MongoDB Visual Guide

Documents, pipelines, replica sets, sharding -- the document database in depth.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

The Document Model

Rich JSON documents replace rows.

```
{ _id: ObjectId("..."), sku: "KB-42",
  name: "Keyboard", price: 129,
  specs: { switches: "tactile", layout: "75%" },
  reviews: [ {user: "ada", stars: 5}, ... ] }
```

Embed

read together:■reviews in product

Reference

grows forever:■orders -> users

One document ~ one read. If a query needs two collections, reconsider the embedding.

STEP 2/6

Aggregation Pipeline

Documents flow through stages, each reshaping the stream.

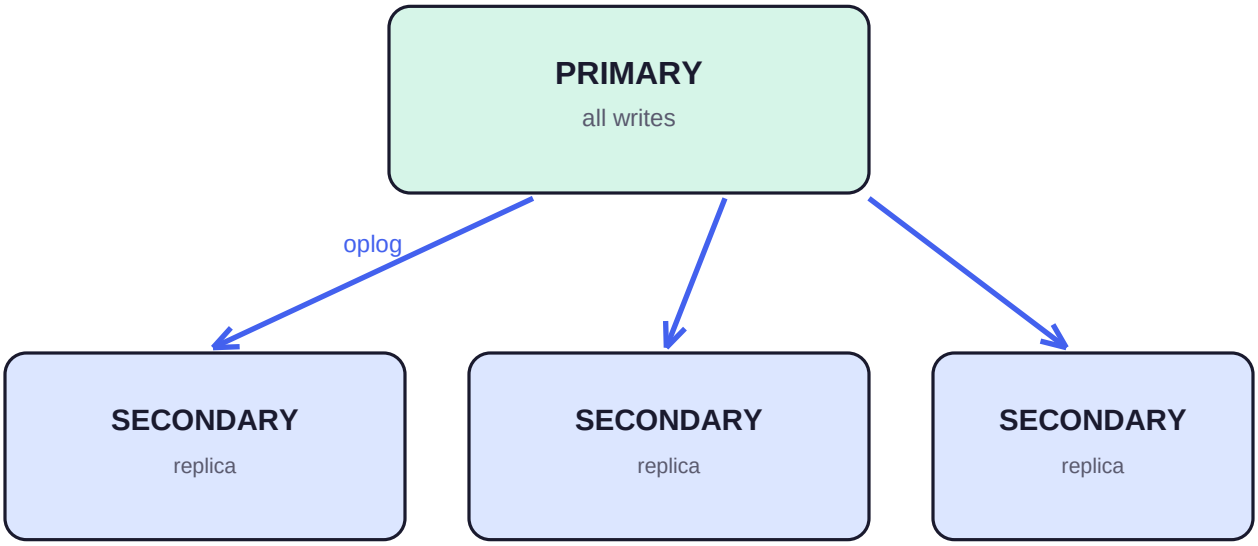
```
db.orders.aggregate([
  { $match:  { status: 'paid' } },
  { $group:  { _id: '$region', total: { $sum: '$amount' } } },
  { $sort:   { total: -1 } },
  { $limit:  5 }
])
```

Pipeline rule: $match early. Filtering first shrinks every downstream stage.

STEP 3/6

Replica Sets

One primary, N secondaries, automatic failover.



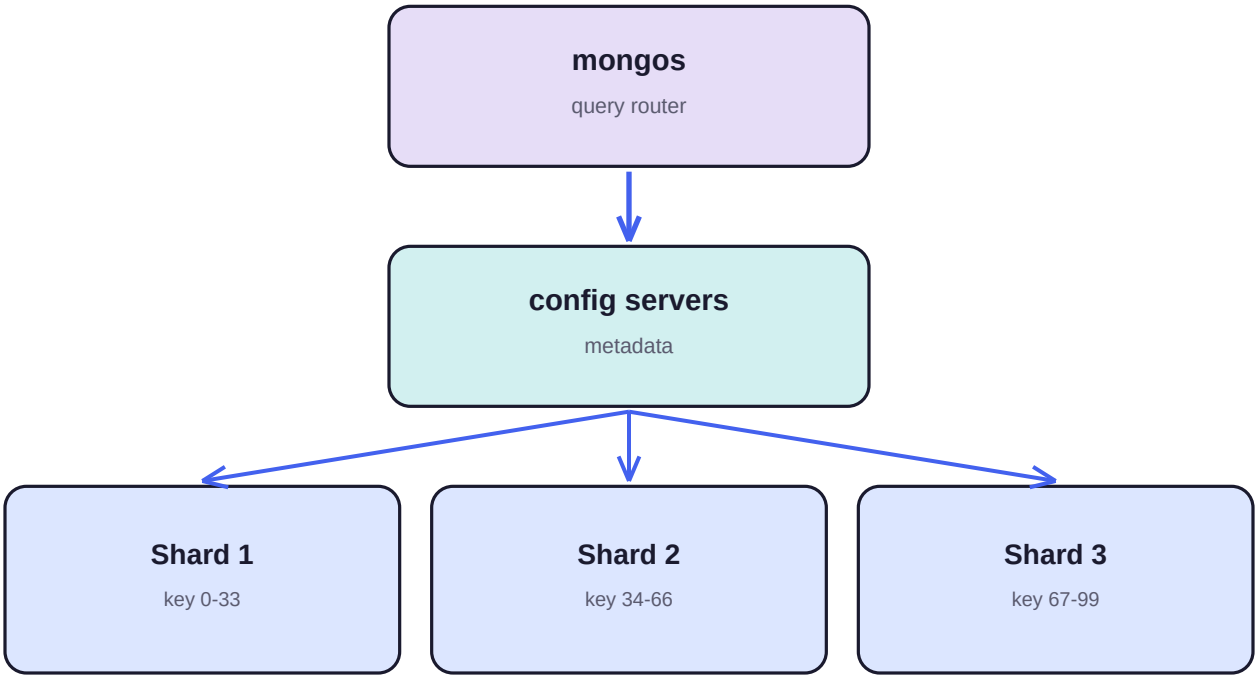
Heartbeat every 2s. Primary down >10s: election, a secondary promotes itself.

Writes go to the primary only. Secondaries are for failover -- and optionally stale-tolerant reads.

STEP 4/6

Sharding

Distribute documents across shards by shard key.



Shard key choice is forever-ish: high cardinality, even distribution, aligned with queries.

STEP 5/6

Consistency Knobs

Tune durability and freshness per operation.

writeConcern

w:1 / w:majority /■w:all + journal

readPreference

primary / secondary /■nearest

```
// strong: acknowledged by a majority, read from primary
db.orders.insertOne(doc, { writeConcern: { w: 'majority' } })
```

Default (w:1, read primary) is fast. Majority writes + primary reads = linearizable-ish safety.

STEP ✓/✓

Key Takeaways

MongoDB -- Remember

- Model as documents: embed what you read together, reference what grows unboundedly.
- Aggregation pipelines reshape streams stage by stage -- always $match early.
- Replica sets give automatic failover via elections; writes go to the primary.
- Sharding scales horizontally but the shard key decision is nearly permanent.
- writeConcern and readPreference let you trade speed for durability per operation.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev